

FUKAURA, Japan

WMO: 475740

Lat: 40.6464N

Lon: 139.9331E

Elev: 223

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	23.9	11.0	9.7	26.8	12.9	10.7	27.6	29.1	31.8	26.2	31.0	14.0	260	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	84.1	75.4	81.8	74.4	79.8	73.4	76.9	81.5	75.8	80.2	74.6	78.8	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	134.5	79.6	74.4	129.5	78.8	73.1	123.9	77.8	40.6	81.8	39.5	80.2	38.4	78.9	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	19.7	17.1	DB	19.8	88.6	1.9	2.5	18.4	90.4	17.3	91.9	16.2	93.3	14.8	95.1	(4)
(5)			WB	18.0	79.0	2.1	1.8	16.5	80.3	15.3	81.3	14.1	82.4	12.6	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	31.7	32.5	37.8	47.3	56.7	63.7	71.4	74.3	67.4	56.8	46.1	36.4		
	DBStd	15.57	4.37	5.16	5.58	6.03	5.43	4.37	4.19	3.89	4.90	5.04	6.40	5.49		
	HDD50	2151	569	491	381	122	8	0	0	0	0	7	150	423		
	HDD65	5399	1034	911	844	531	263	74	2	0	27	258	568	887		
	CDD50	2862	0	0	2	41	214	412	665	755	523	217	32	1		
	CDD65	635	0	0	0	0	5	36	202	290	100	2	0	0		
	CDH74	2849	0	0	0	0	11	51	793	1639	354	1	0	0		
	CDH80	414	0	0	0	0	0	1	96	267	50	0	0	0		
	WSAvg	7.3	10.0	9.6	8.9	7.7	6.5	5.0	5.1	5.0	5.5	6.7	8.3	9.9		
	PrecAvg	59.5	4.0	3.2	3.2	3.6	4.7	3.7	5.9	6.5	6.7	6.4	6.3	5.4		
	PrecMax	74.7	8.0	5.8	6.2	5.9	8.6	8.5	13.3	12.7	16.6	14.0	15.4	9.1		
(15) Precipitation	PrecMin	43.2	1.9	1.1	1.2	1.1	2.4	0.4	1.3	2.2	2.0	2.2	3.3	2.1		
	PrecStd	8.4	1.4	1.3	1.1	1.3	1.7	2.0	2.8	2.8	3.2	2.5	2.2	1.8		
	DB	47.4	52.1	57.9	68.5	75.4	78.4	86.0	87.9	84.7	72.2	65.0	55.7			
	MCWB	42.4	46.0	46.3	54.7	60.8	67.6	76.3	76.8	74.5	64.8	57.0	51.0			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	42.6	46.5	53.4	64.1	71.5	75.3	82.5	85.2	81.0	69.5	61.6	51.0			
	2% MCWB	38.8	41.7	46.3	53.3	59.3	66.1	74.3	76.2	73.2	62.3	55.6	47.1			
	5% DB	40.0	42.8	50.1	60.0	68.2	73.1	80.3	82.8	77.8	67.3	59.0	47.6			
	5% MCWB	36.2	38.5	44.7	50.5	58.4	65.4	73.2	75.3	71.0	60.8	53.7	43.7			
	10% DB	38.0	40.0	47.0	56.7	65.5	71.1	78.2	81.0	75.4	65.1	56.3	44.7			
	10% MCWB	34.2	35.8	42.2	49.2	57.4	64.8	72.4	74.5	69.4	59.1	51.2	40.5			
	0.4% WB	43.5	47.3	50.3	58.1	63.8	70.6	77.4	78.8	76.5	67.9	59.8	51.7			
	0.4% MCDB	46.5	50.6	54.2	64.2	69.9	74.6	82.7	84.0	82.1	70.8	62.5	53.6			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.3	42.0	47.7	54.7	61.9	68.8	76.0	77.5	74.1	64.2	56.9	47.6			
	2% MCDB	41.9	45.5	51.6	60.9	67.4	72.5	80.6	82.2	79.1	67.3	60.1	50.3			
	5% WB	36.8	38.8	45.2	52.3	60.3	67.3	74.6	76.5	72.3	62.0	54.3	44.0			
	5% MCDB	39.5	42.4	49.3	57.5	65.9	71.1	78.8	80.9	76.7	66.0	58.3	47.3			
	10% WB	34.6	36.2	42.5	50.3	58.6	65.9	73.4	75.5	70.4	60.0	51.7	40.8			
	10% MCDB	37.5	39.7	46.6	55.3	64.0	69.7	77.0	79.8	74.5	64.0	55.6	44.1			
(35) Mean Daily Temperature Range	MDBR	7.0	8.0	10.3	13.0	13.0	11.7	10.2	11.3	12.4	12.1	10.2	8.3			
	5% DB	10.7	12.6	15.4	16.7	16.6	14.5	12.5	12.8	14.0	14.1	14.0	13.3			
	5% WB	9.5	10.9	12.1	10.6	9.9	8.8	7.1	7.4	8.4	9.9	12.0	12.0			
	MCDBR	10.6	12.5	14.8	14.3	14.4	11.9	10.4	10.9	12.5	12.6	13.6	13.1			
	MCWBR	10.1	11.7	12.6	10.6	9.9	7.9	6.3	6.8	8.3	10.2	12.9	12.7			
(40) Clear-Sky Solar Irradiance	taub	0.383	0.425	0.493	0.538	0.553	0.525	0.534	0.493	0.435	0.419	0.412	0.397			
	taud	2.055	1.951	1.852	1.804	1.848	2.004	2.051	2.180	2.294	2.198	2.107	2.044			
	Ebn at Noon	231	239	237	236	237	244	240	247	254	241	220	213			
	Edh at Noon	37	48	59	66	64	55	52	45	37	37	35	35			
(44) All-Sky Solar Radiation	RadAvg	390	641	1047	1463	1610	1634	1449	1482	1231	912	518	354			
	RadStd	31	71	84	149	168	149	147	144	99	61	51	34			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page